

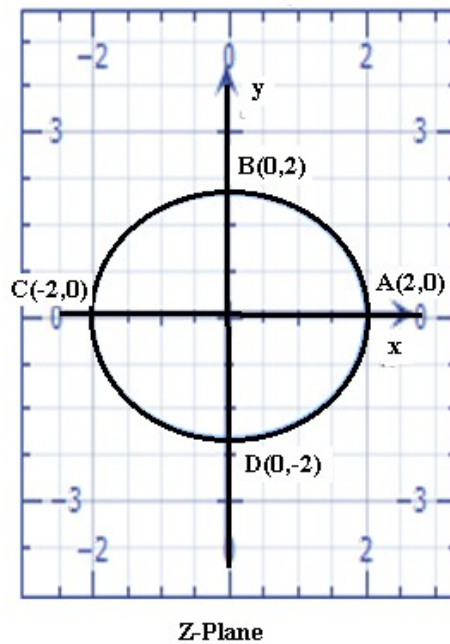
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Assignment on

Mathematics-V

Home Task-1: Determine the image in the w-plane of the circle $|z|=2$ in the z-plane under the transformation $w=\frac{z+j}{z-j}$.



From figure, $OP = |z| = \sqrt{x^2 + y^2}$

Given,

$$|z| = 2$$

$$\sqrt{x^2 + y^2} = 2$$

$$x^2 + y^2 = 2^2 \quad [\text{Squaring}]$$

$$\therefore x^2 + y^2 = 4$$

$$(x-0)^2 + (y-0)^2 = 2^2 \text{ -----(i)}$$

[We have, $(x-a)^2 + (y-b)^2 = r^2$]

The equation (i) represents a circle whose Center (0, 0) & radius = 2

Given,

$$w = \frac{z+j}{z-j}$$

$$w = \frac{x+jy+j}{x+jy-j}$$

$$\Rightarrow u+jv = \frac{x+j(y+1)}{x+j(y-1)}$$

$$\Rightarrow u+jv = \frac{\{x+j(y+1)\}\{x-j(y-1)\}}{\{x+j(y-1)\}\{x-j(y-1)\}}$$

$$\Rightarrow u + jv = \frac{(x+jy+j)(x-jy+)}{x^2-(j(y-1))^2}$$

$$\Rightarrow u + jv = \frac{x^2 - xjy + xj + jxy - j^2y^2 + j^2y + jx - j^2y + j^2}{x^2 + (y-1)^2}$$

$$\Rightarrow u + jv = \frac{x^2 + y^2 + 2jx - 1}{x^2 + (y-1)^2}$$

$$\Rightarrow u + jv = \frac{x^2 + y^2 - 1}{x^2 + y^2 - 2y + 1} + \frac{j2x}{x^2 + y^2 - 2y + 1}$$

Equating the real and imaginary part we get,

$$u = \frac{x^2 + y^2 - 1}{x^2 + y^2 - 2y + 1} \quad \text{ii}$$

$$v = \frac{2x}{x^2 + y^2 - 2y + 1} \quad \text{iii}$$

we have ,

$$\therefore x^2 + y^2 = 4$$

Putting the value of $x^2 + y^2 = 4$ in equation ii & iii we get,

$$u = \frac{x^2 + y^2 - 1}{x^2 + y^2 - 2y + 1} \quad v = \frac{2x}{x^2 + y^2 - 2y + 1}$$

$$\Rightarrow u = \frac{4-1}{4-2y+1} \quad v = \frac{2x}{4-2y+1}$$

$$u = \frac{3}{5-2y} \quad v = \frac{2x}{5-2y}$$

$$5 - 2y = \frac{3}{u}$$

$$5 - \frac{3}{u} = 2y$$

$$2y = \frac{5u-3}{u}$$

$$y = \frac{5u-3}{2u}$$

$$v = \frac{2x}{5-2y}$$

$$2x = v(5-2y)$$

$$x = \frac{v\{5-2(\frac{5u-3}{2u})\}}{2}$$

$$= \frac{v\{5-(\frac{10u-6}{2u})\}}{2}$$

$$= \frac{v\{(\frac{10u-10u+6}{2u})\}}{2}$$

$$= \frac{v\{(\frac{6}{2u})\}}{2}$$

$$= \frac{\frac{3v}{u}}{2}$$

$$\therefore x = \frac{3v}{2u}$$

Putting the value of x and y in equation we get,

$$x^2 + y^2 = 4$$

$$(\frac{3v}{2u})^2 + (\frac{5u-3}{2u})^2 - 4 = 0.$$

$$\Rightarrow \frac{9v^2}{4u^2} + (\frac{5u-3}{2u})^2 - 4 = 0$$

$$\Rightarrow \frac{9v^2}{4u^2} + \frac{25u^2-30u+9}{4u^2} - 4 = 0$$

$$\frac{9v^2+25u^2-30u+9-16u^2}{4u^2} = 0$$

$$\frac{9v^2+9u^2-30u+9}{4u^2} = 0$$

$$\Rightarrow 9u^2 + 9v^2 - 30u + 9 = 0$$

$$\Rightarrow u^2 + v^2 - \frac{30u}{9} + 1 = 0$$

$$\Rightarrow u^2 + v^2 - 2 \cdot \frac{15}{9}u + 2 \cdot 0 \cdot v + 1 = 0$$

$$\Rightarrow u^2 + v^2 + 2(-\frac{15}{9})u + 2 \cdot 0 \cdot v + 1 = 0$$

$$\Rightarrow u^2 + v^2 + 2(-\frac{5}{3})u + 2 \cdot 0 \cdot v + 1 = 0$$

We know the general equation of circle is,

$$x^2 + y^2 + 2gx + 2fy + c = 0.$$

From equation the centre is $(-g, -f) = (\frac{5}{3}, 0)$

$$\text{Radius} = \sqrt{g^2 + f^2 - c}$$

$$\text{Radius is } \sqrt{g^2 + f^2 - c} = \sqrt{(-\frac{5}{3})^2 + 0^2 - 1}$$

$$= \sqrt{\frac{25}{9} + 0 - 1}$$

$$= \sqrt{\frac{25 - 9}{9}}$$

$$= \sqrt{\frac{16}{9}}$$

$$= \frac{4}{3}$$

$\therefore$ The centre of the new circle in the w plane is $D(\frac{5}{3}, 0)$

now Taking three sample point A(2,0), B(0,2), C (-2,0)

We get from equation 6 and 7

For A(2,0)

$$U = \frac{3}{5 - 2 \cdot 0}$$

$$V = \frac{2 \cdot 2}{5 - 2 \cdot 0} = \frac{4}{5}$$

$$= \frac{3}{5}$$

$\therefore$ the image of A is $A'(\frac{3}{5}, \frac{4}{5})$

For B(0,2)

$$U = \frac{3}{5-2.2} = 3$$

$$V = \frac{2.0}{5-2.2} = 0$$

∴ the image of B is B'(3,0)

For c (-2,0)

$$U = \frac{3}{5-2.0} = \frac{3}{5}$$

$$V = \frac{32 \cdot (-2)}{5-2.0} = -\frac{4}{5}$$

∴ the image of C is C'($\frac{3}{5}, -\frac{4}{5}$).